

ROI Analyses Hippocampus - Model 1 (Encode & Recall)

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Hippocampus")
ROIdata <- read.csv("BOLD_Hipp_model1.csv")

# Assign factors
ROIdata$S_ID <- factor(ROIdata$S_ID)
ROIdata$ID <- factor(ROIdata$ID)
ROIdata$ROI <- factor(ROIdata$ROI)
ROIdata$hemisphere <- factor(ROIdata$hemisphere)
ROIdata$StimType <- factor(ROIdata$StimType)
ROIdata$TaskStage <- factor(ROIdata$TaskStage)
```

Sham Condition - Hippocampus - Figure 2F

```
# Does the hippocampus show different activity between encode and recall stages of the task?
# BOLD = median % BOLD signal

Hip_data_Shram <- subset(ROIdata, ROI=="Hippocampus" & StimType=="Sham")

# One sample t-test
upper = 0.3
type = "two.sided"
Sham_ost <- ddp1y(Hip_data_Shram, .(TaskStage, hemisphere), t.test.p1yr, "BOLD")
Sham_ost <- ost_FDR(Sham_ost)
pander(Sham_ost)
```

Table 1: Table continues below

TaskStage	hemisphere	statistic	p.value
Encode	L	3.69812064228076	0.00152626851940014

TaskStage	hemisphere	statistic	p.value
Encode	R	3.9214807605266	0.000916710999101395
Recall	L	-1.28120701566625	0.215538142368785
Recall	R	-0.250312947945414	0.805032026471034

Table 2: Table continues below

estimate	conf.int1	conf.int2	nobs	dof
0.13757115	0.0597100540555932	0.215432245944407	20	19
0.15196645	0.070856926649865	0.233075973350135	20	19
-0.0593915	-0.156415506743164	0.0376325067431636	20	19
-0.015651	-0.146518858592309	0.115216858592309	20	19

method	alternative	null.value	sig	FDR
One Sample t-test	two.sided	0	**	0.003053
One Sample t-test	two.sided	0	**	0.003053
One Sample t-test	two.sided	0	NA	0.2874
One Sample t-test	two.sided	0	NA	0.805

LMM - effect of task stage and hemisphere on BOLD signal

```
BA <- lmer(BOLD ~ TaskStage*hemisphere + (1|S_ID), data = Hip_data_Shame, REML = TRUE)
Anova(BA, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
```

```
##
```

```
## Response: BOLD
```

```
##              F Df Df.res    Pr(>F)
## TaskStage    20.4921  1    57 3.097e-05 ***
## hemisphere    0.5211  1    57  0.4733
## TaskStage:hemisphere 0.1328  1    57  0.7169
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(BA, ddf = "Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
```

```
## lmerModLmerTest]
```

```
## Formula: BOLD ~ TaskStage * hemisphere + (1 | S_ID)
```

```
## Data: Hip_data_Shame
```

```
##
```

```
## REML criterion at convergence: -15.4
```

```
##
```

```
## Scaled residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -2.93762 -0.45896 -0.00748  0.39611  2.48798
```

```
##
```

```
## Random effects:
```

```
## Groups   Name      Variance Std.Dev.
```

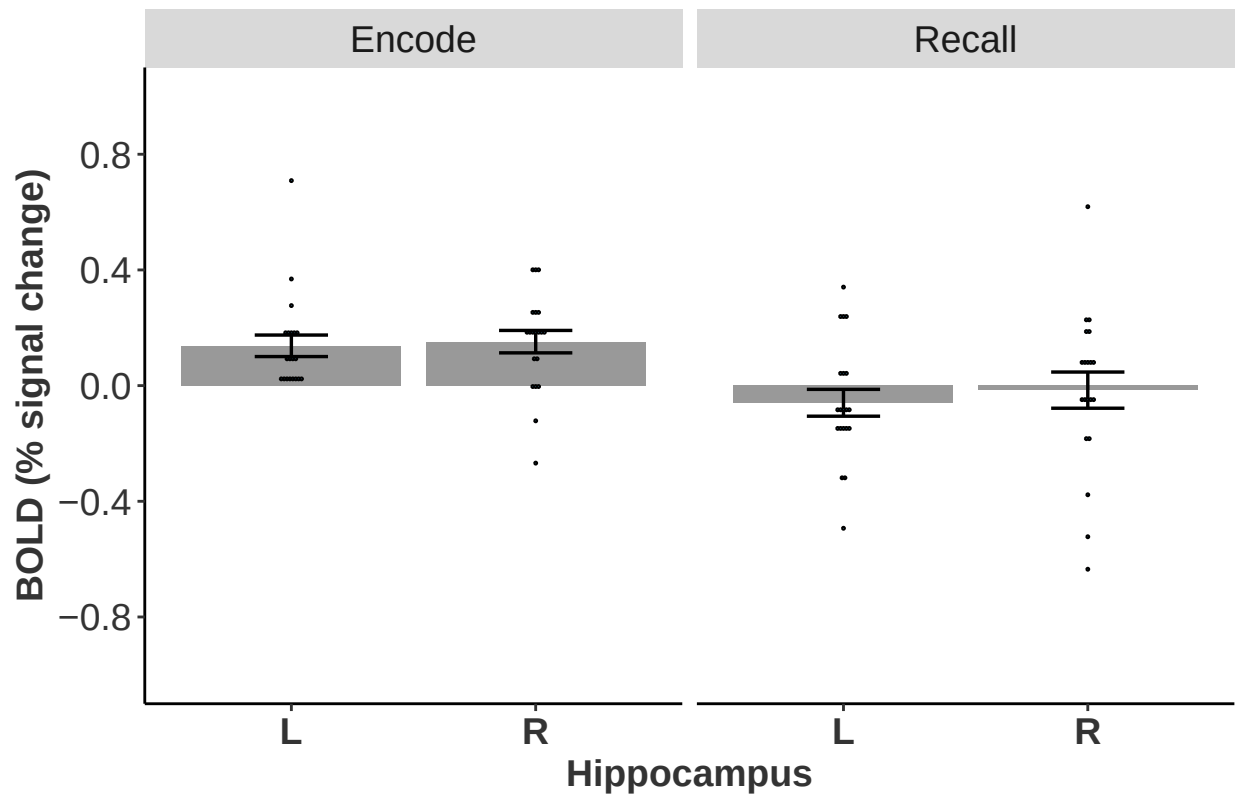
```
## S_ID      (Intercept) 0.01229  0.1109
## Residual          0.03243  0.1801
## Number of obs: 80, groups: S_ID, 20
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.13757    0.04729 61.96442   2.909  0.00502 **
## TaskStageRecall -0.19696    0.05695 57.00000  -3.459  0.00103 **
## hemisphereR     0.01440    0.05695 57.00000   0.253  0.80135
## TaskStageRecall:hemisphereR 0.02935    0.08054 57.00000   0.364  0.71693
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) TskStR hmsphR
## TaskStgRc11 -0.602
## hemisphereR -0.602  0.500
## TskStgRcl:R  0.426 -0.707 -0.707
```

```
estimated_means <- emmeans(BA, ~ TaskStage*hemisphere)
estimated_means
```

```
## TaskStage hemisphere emmean      SE df lower.CL upper.CL
## Encode      L         0.1376 0.0473 62   0.0430   0.2321
## Recall      L        -0.0594 0.0473 62  -0.1539   0.0351
## Encode      R         0.1520 0.0473 62   0.0574   0.2465
## Recall      R        -0.0157 0.0473 62  -0.1102   0.0789
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
# Plot
p.Sham_hipp <-ggplot(Hip_data_Sham, aes(x = hemisphere, y = BOLD, fill=hemisphere)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, .8, .4),
    limits=c(-1, 1)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    panel.background = element_blank()) +
  theme(legend.title=element_blank(), legend.position="none") +
  scale_colour_manual(values=c("#999999", "#999999"))+
  scale_fill_manual(values=c("#999999", "#999999")) +
  scale_x_discrete(breaks=c("L","R"),
    labels=c("L", "R")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("") + xlab("Hippocampus")
```

```
ggsave(p.Sham_hipp ,file="Figure_2F.pdf", width = 9, height = 9, dpi = 300, units = "cm", device = "pdf")
p.Sham_hipp
```



Sham Condition - segmented Hippocampus - Figure 2G

```
## Does the left hippocampus shows different activity during encode across hippocampal regions (anterior, middle, posterior)

Left_Seg_Hip_data_Shame <- subset(ROIdata, (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & StimType=="Sham" & Condition=="Encode")
BA_seg <- lmer(BOLD ~ ROI + (1|S_ID), data = Left_Seg_Hip_data_Shame, REML = TRUE)
Anova(BA_seg, test="F")

## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##          F Df Df.res    Pr(>F)
## ROI 8.7171  2     38 0.0007658 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(BA_seg, ddf = "Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: BOLD ~ ROI + (1 | S_ID)
## Data: Left_Seg_Hip_data_Sham
##
## REML criterion at convergence: -46.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8325 -0.5095 -0.1110  0.4632  1.9199
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.02966 0.1722
## Residual 0.01034 0.1017
## Number of obs: 60, groups: S_ID, 20
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 0.20153 0.04472 27.14422 4.506 0.000114 ***
## ROIMid -0.12938 0.03215 38.00000 -4.024 0.000263 ***
## ROIPost -0.09569 0.03215 38.00000 -2.976 0.005052 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) ROIMid
## ROIMid -0.359
## ROIPost -0.359 0.500
```

```
BA_contrasts <- emmeans(BA_seg, specs = pairwise ~ ROI, type = "response")
BA_contrasts
```

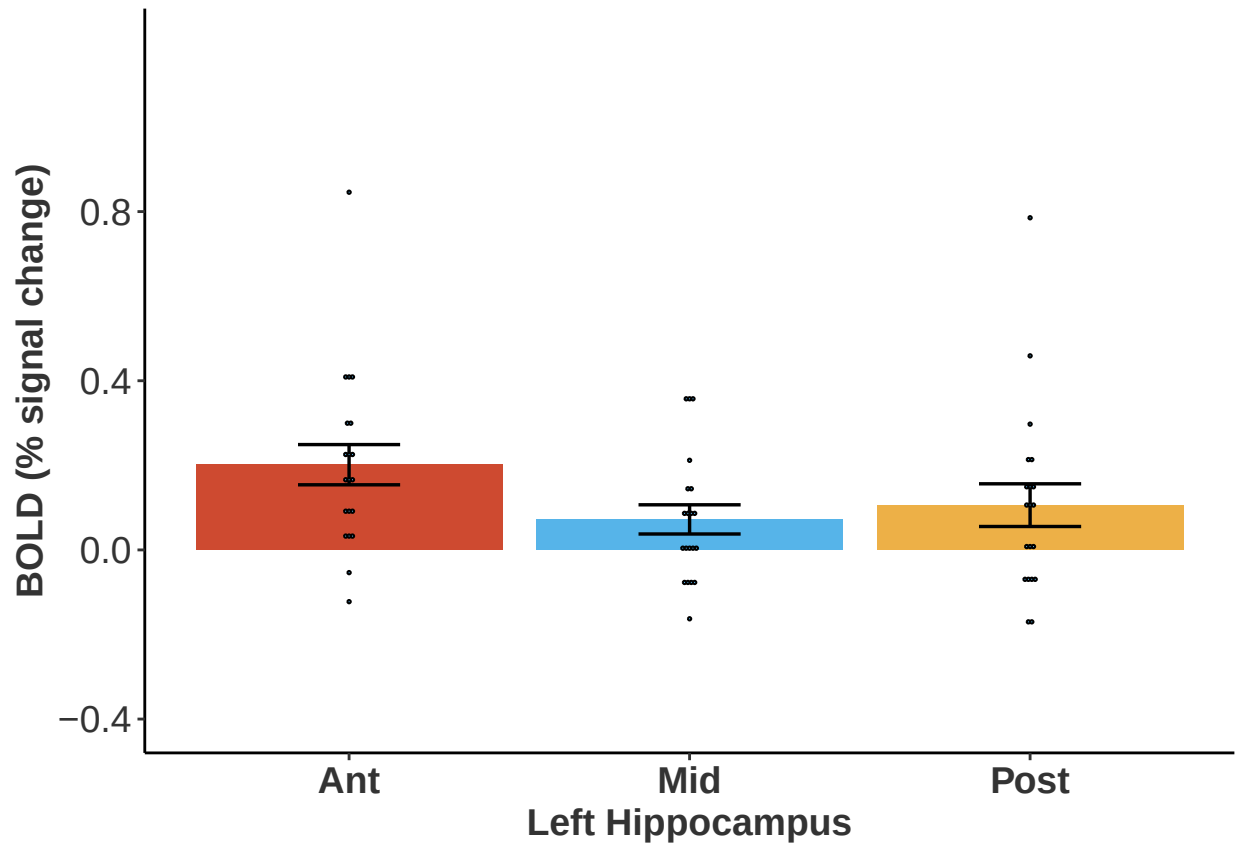
```
## $emmeans
## ROI emmean SE df lower.CL upper.CL
## Ant 0.2015 0.0447 27.1 0.1098 0.293
## Mid 0.0721 0.0447 27.1 -0.0196 0.164
## Post 0.1058 0.0447 27.1 0.0141 0.198
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## Ant - Mid 0.1294 0.0322 38 4.024 0.0008
## Ant - Post 0.0957 0.0322 38 2.976 0.0137
## Mid - Post -0.0337 0.0322 38 -1.048 0.5518
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```

# Create plot
p.BA_LHipp_Sham <- ggplot(Left_Seg_Hip_data_Sham, aes(x = ROI, y = BOLD, fill = ROI)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.5) +
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.4, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  scale_x_discrete(breaks=c("Ant","Mid", "Post"),
    labels=c("Ant","Mid", "Post")) +
  scale_fill_manual(values=c("#ce4a30", "#56B4E9","#edb047")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + xlab("Left Hippocampus")

ggsave(p.BA_LHipp_Sham ,file="Figure_2G.pdf", width = 7, height = 8, dpi = 300, units = "cm", device = "pdf")
p.BA_LHipp_Sham

```



What about the right hippocampus?

```
Right_Seg_Hip_data_Sham <- subset(ROIdata, (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & StimType=="Sham" &
BA_seg <- lmer(BOLD ~ ROI + (1|S_ID), data = Right_Seg_Hip_data_Sham, REML = TRUE)
Anova(BA_seg, test="F")
```

Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)

##

Response: BOLD

F Df Df.res Pr(>F)

ROI 0.1991 2 38 0.8203

no ROI effect for the right hippocampus

Stim Conditions - whole Hippocampus - Figures 3B and 3G

Does stimulation impact BOLD activity in the targeted hippocampus - Left ?

```
Hip_data_L<- subset(ROIdata, ROI=="Hippocampus" & hemisphere=="L")
```

```
BA <- lmer(BOLD ~ StimType*TaskStage + (1|S_ID), data = Hip_data_L, REML = TRUE)
```

```
Anova(BA, test="F")
```

Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)

##

```
## Response: BOLD
##               F Df Df.res    Pr(>F)
## StimType      3.2224  2    95  0.04425 *
## TaskStage     44.8436  1    95 1.492e-09 ***
## StimType:TaskStage 2.9611  2    95  0.05656 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(BA, ddf = "Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: BOLD ~ StimType * TaskStage + (1 | S_ID)
## Data: Hip_data_L
##
## REML criterion at convergence: -81.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.53487 -0.51943 -0.00584  0.58444  2.95468
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.01817  0.1348
## Residual 0.01767  0.1329
## Number of obs: 120, groups: S_ID, 20
##
## Fixed effects:
##               Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)      0.13757    0.04233 49.88460   3.250  0.00207
## StimTypeTI 1:1    -0.01244    0.04203 95.00000  -0.296  0.76792
## StimTypeTI 1:3    -0.13167    0.04203 95.00000  -3.133  0.00230
## TaskStageRecall   -0.19696    0.04203 95.00000  -4.686 9.29e-06
## StimTypeTI 1:1:TaskStageRecall -0.01420    0.05944 95.00000  -0.239  0.81175
## StimTypeTI 1:3:TaskStageRecall  0.11757    0.05944 95.00000   1.978  0.05084
##
## (Intercept)          **
## StimTypeTI 1:1
## StimTypeTI 1:3          **
## TaskStageRecall       ***
## StimTypeTI 1:1:TaskStageRecall
## StimTypeTI 1:3:TaskStageRecall .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) StTTI1:1 StTTI1:3 TskStR STTI1:1:
## StmTypTI1:1 -0.496
## StmTypTI1:3 -0.496  0.500
## TaskStgRc1l -0.496  0.500  0.500
## STTI1:1:TSR  0.351 -0.707 -0.354 -0.707
## STTI1:3:TSR  0.351 -0.354 -0.707 -0.707  0.500
```

```
BA_contrasts <- emmeans(BA, specs = pairwise ~ StimType|TaskStage, type = "response")
BA_contrasts
```

```
## $emmeans
## TaskStage = Encode:
##   StimType   emmean      SE    df lower.CL upper.CL
##   Sham       0.1376 0.0423 49.9    0.0525    0.2226
##   TI 1:1      0.1251 0.0423 49.9    0.0401    0.2102
##   TI 1:3      0.0059 0.0423 49.9   -0.0791    0.0909
##
## TaskStage = Recall:
##   StimType   emmean      SE    df lower.CL upper.CL
##   Sham      -0.0594 0.0423 49.9   -0.1444    0.0256
##   TI 1:1     -0.0860 0.0423 49.9   -0.1711   -0.0010
##   TI 1:3     -0.0735 0.0423 49.9   -0.1585    0.0115
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## TaskStage = Encode:
##   contrast      estimate      SE df t.ratio p.value
##   Sham - TI 1:1      0.0124 0.042 95    0.296  0.9529
##   Sham - TI 1:3      0.1317 0.042 95    3.133  0.0065
##   TI 1:1 - TI 1:3     0.1192 0.042 95    2.837  0.0153
##
## TaskStage = Recall:
##   contrast      estimate      SE df t.ratio p.value
##   Sham - TI 1:1      0.0266 0.042 95    0.634  0.8020
##   Sham - TI 1:3      0.0141 0.042 95    0.335  0.9399
##   TI 1:1 - TI 1:3    -0.0125 0.042 95   -0.298  0.9522
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Plot - Figure 2K

```
p.BA.hipp_L <- ggplot(Hip_data_L, aes(x = StimType, y = BOLD, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.6, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1)
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
```

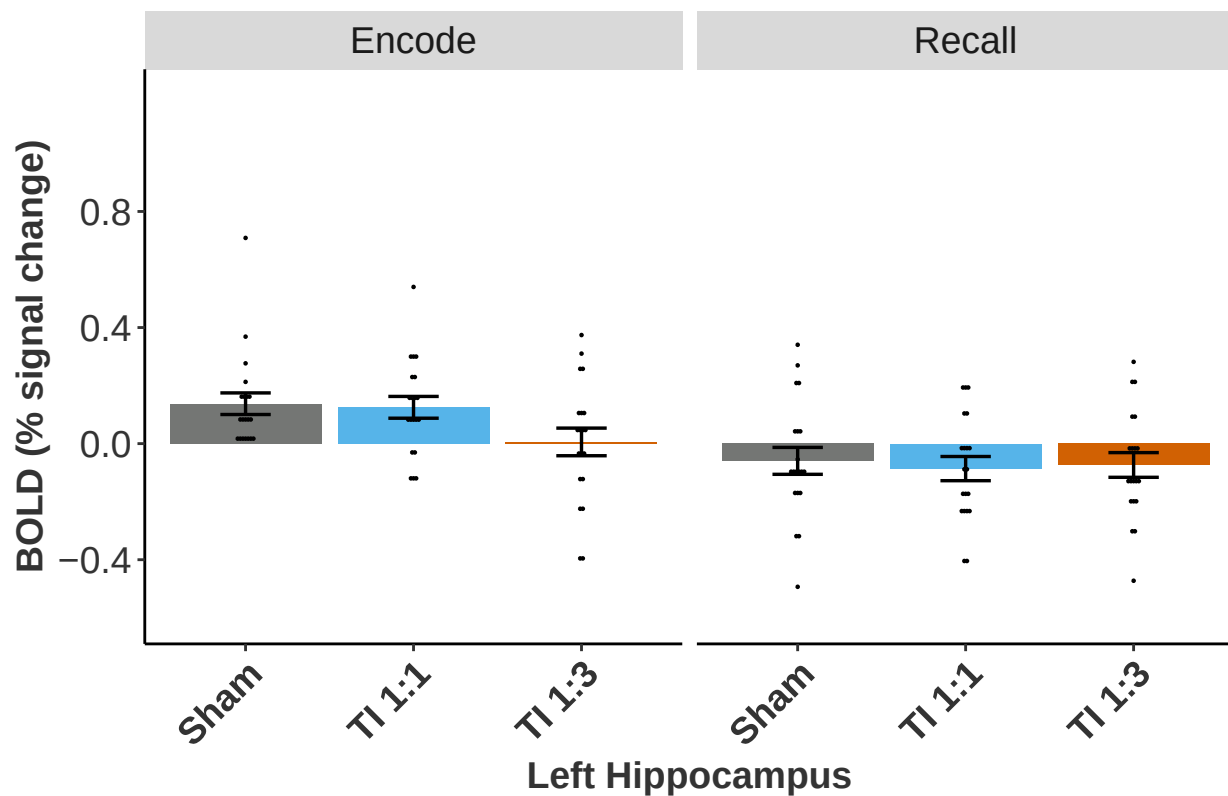
```

scale_x_discrete(breaks=c("Sham","TI 1:1","TI 1:3"),
labels=c("Sham","TI 1:1","TI 1:3")) +
theme(panel.grid.major = element_blank(),
panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
theme(strip.text.x = element_text(size = 14),
strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("") + xlab("Left Hippocampus")

ggsave(p.BA.hipp_L,file="Figure_3B.pdf", width = 9, height = 9, dpi = 300, units = "cm", device = "pdf")

p.BA.hipp_L

```



```

# What about the right hippocampus, not targeted?

Hip_data_R<- subset(ROIdata, ROI=="Hippocampus" & hemisphere=="R")
BA <- lmer(BOLD ~ StimType*TaskStage + (1|S_ID), data = Hip_data_R, REML = TRUE)
Anova(BA, test="F")

## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##
##          F Df Df.res    Pr(>F)
## StimType  0.7244  2    95   0.4873
## TaskStage 18.3715  1    95 4.359e-05 ***

```

```

## StimType:TaskStage 2.0744 2 95 0.1313
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(BA, ddf = "Kenward-Roger")

## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: BOLD ~ StimType * TaskStage + (1 | S_ID)
## Data: Hip_data_R
##
## REML criterion at convergence: -27.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.48440 -0.49093  0.06146  0.55882  2.70726
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.02132 0.1460
## Residual 0.02970 0.1723
## Number of obs: 120, groups: S_ID, 20
##
## Fixed effects:
##
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 0.15197 0.05051 60.86466 3.009 0.00381 **
## StimTypeTI 1:1 -0.01979 0.05450 95.00000 -0.363 0.71727
## StimTypeTI 1:3 -0.10629 0.05450 95.00000 -1.950 0.05409 .
## TaskStageRecall -0.16762 0.05450 95.00000 -3.076 0.00274 **
## StimTypeTI 1:1:TaskStageRecall -0.02407 0.07708 95.00000 -0.312 0.75549
## StimTypeTI 1:3:TaskStageRecall 0.12231 0.07708 95.00000 1.587 0.11585
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) StTTI1:1 StTTI1:3 TskStR STTI1:1:
## StmTypTI1:1 -0.540
## StmTypTI1:3 -0.540 0.500
## TaskStgRc11 -0.540 0.500 0.500
## STTI1:1:TSR 0.381 -0.707 -0.354 -0.707
## STTI1:3:TSR 0.381 -0.354 -0.707 -0.707 0.500

BA_contrasts <- emmeans(BA, specs = pairwise ~ StimType|TaskStage, type = "response")
BA_contrasts

## $emmeans
## TaskStage = Encode:
## StimType emmean SE df lower.CL upper.CL
## Sham 0.15197 0.0505 60.9 0.0510 0.2530
## TI 1:1 0.13217 0.0505 60.9 0.0312 0.2332
## TI 1:3 0.04567 0.0505 60.9 -0.0553 0.1467
##
## TaskStage = Recall:

```

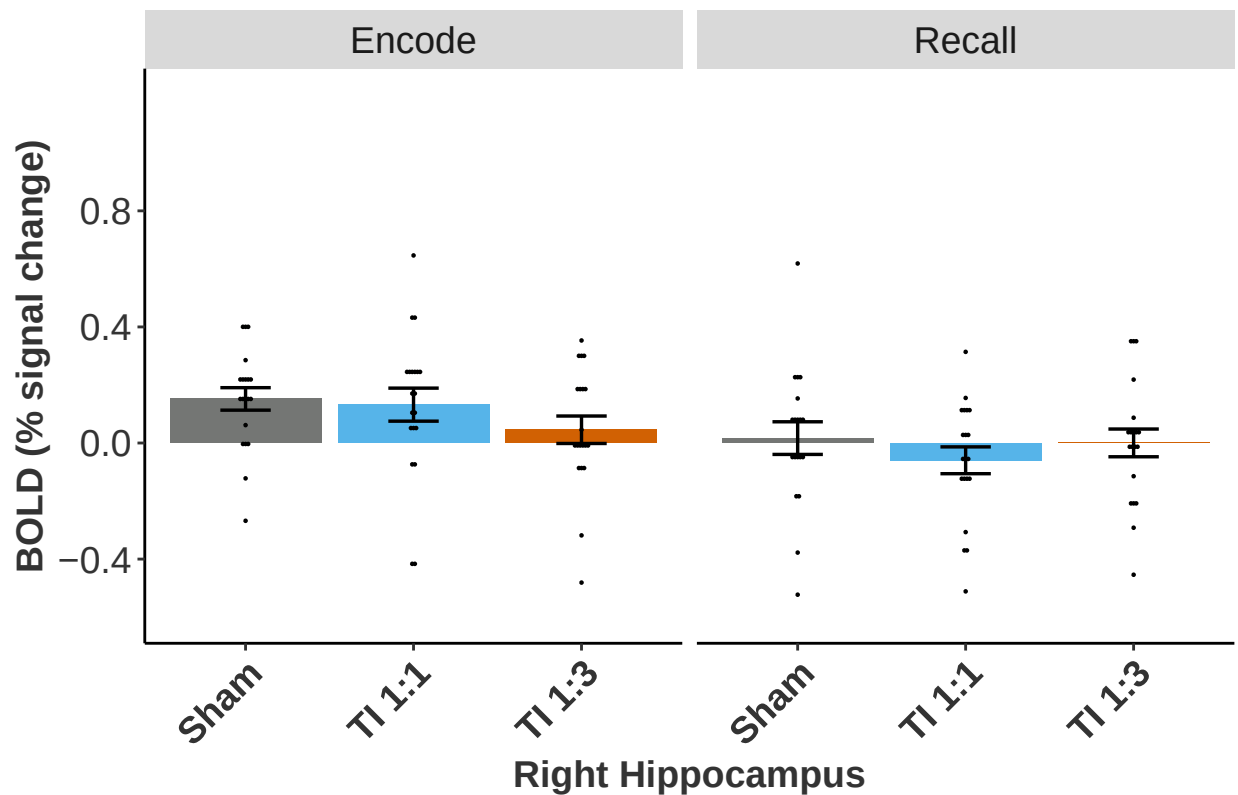
```
## StimType      emmean      SE    df lower.CL upper.CL
## Sham         -0.01565 0.0505 60.9 -0.1167  0.0854
## TI 1:1       -0.05952 0.0505 60.9 -0.1605  0.0415
## TI 1:3        0.00037 0.0505 60.9 -0.1006  0.1014
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## TaskStage = Encode:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:1    0.0198 0.0545 95   0.363  0.9299
## Sham - TI 1:3    0.1063 0.0545 95   1.950  0.1304
## TI 1:1 - TI 1:3  0.0865 0.0545 95   1.587  0.2562
##
## TaskStage = Recall:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:1    0.0439 0.0545 95   0.805  0.7009
## Sham - TI 1:3   -0.0160 0.0545 95  -0.294  0.9535
## TI 1:1 - TI 1:3 -0.0599 0.0545 95  -1.099  0.5172
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Plot - Figure 4L

```
p.BA.hipp_R <- ggplot(Hip_data_R, aes(x = StimType, y = BOLD, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.6, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Sham","TI 1:1","TI 1:3"),
    labels=c("Sham","TI 1:1","TI 1:3")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("") + xlab("Right

ggsave(p.BA.hipp_R,file="Figure_3G.pdf", width = 9, height = 9, dpi = 300, units = "cm", device = "pdf")

p.BA.hipp_R
```



Stim Conditions - segmented Left Hippocampus - Figure 3C

Effect of stimulation was seen only for the left hippocampus - Does it affect segments of the hippocampus?

```
Seg_Hip_data_enc_L <- subset(ROIdata, (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & TaskStage=="Encode" & 1
BA_seg <- lmer(BOLD ~ StimType*ROI + (1|S_ID), data = Seg_Hip_data_enc_L, REML = TRUE)
Anova(BA_seg, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
```

```
##
```

```
## Response: BOLD
```

```
##           F Df Df.res    Pr(>F)
## StimType 12.6459 2    152 8.313e-06 ***
## ROI       6.3506 2    152 0.002245 **
## StimType:ROI 0.4639 4    152 0.762166
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(BA_seg, ddf = "Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
```

```
## lmerModLmerTest]
```

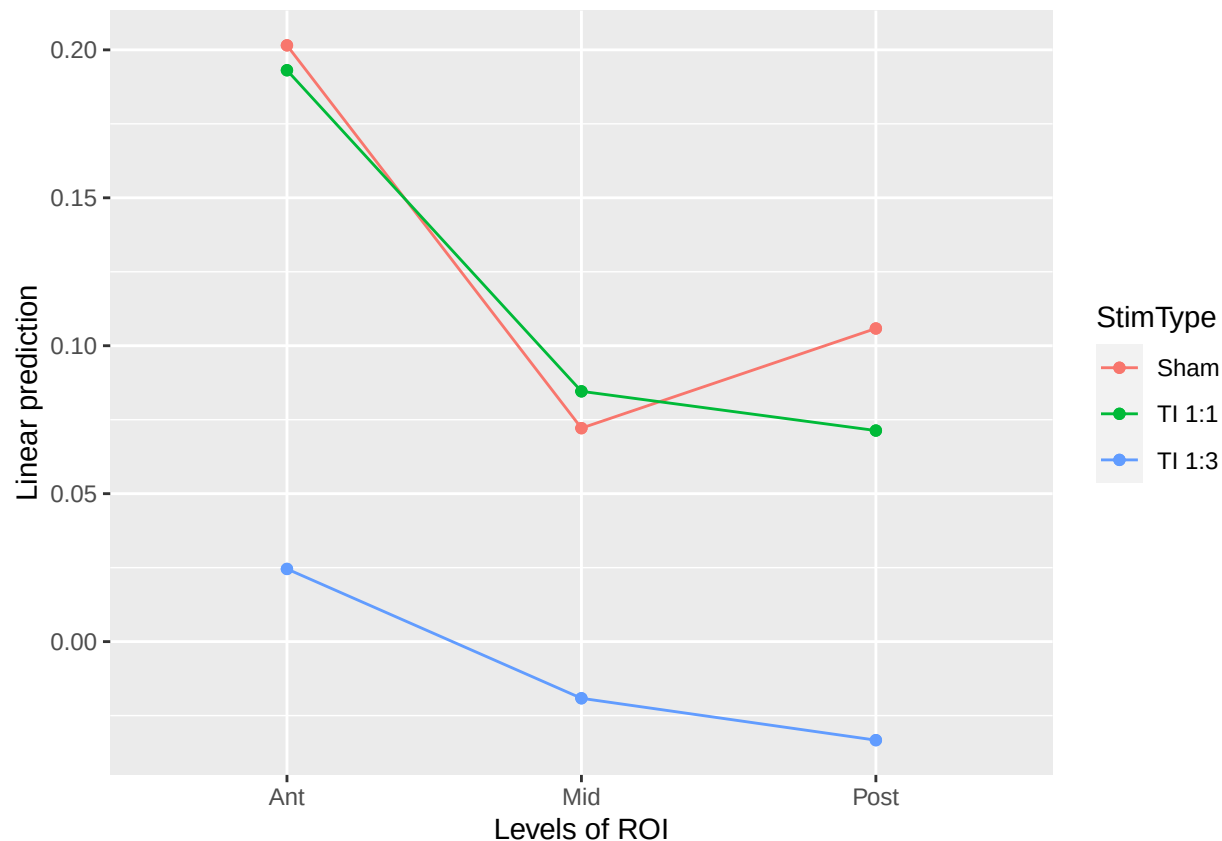
```
## Formula: BOLD ~ StimType * ROI + (1 | S_ID)
```

```

## Data: Seg_Hip_data_enc_L
##
## REML criterion at convergence: -63.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.84114 -0.67097  0.02941  0.56009  2.57765
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.02302 0.1517
## Residual 0.02715 0.1648
## Number of obs: 180, groups: S_ID, 20
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 0.201525 0.050085 63.693682 4.024 0.000155 ***
## StimTypeTI 1:1 -0.008413 0.052103 152.000000 -0.161 0.871932
## StimTypeTI 1:3 -0.176966 0.052103 152.000000 -3.396 0.000871 ***
## ROIMid -0.129385 0.052103 152.000000 -2.483 0.014106 *
## ROIPost -0.095693 0.052103 152.000000 -1.837 0.068222 .
## StimTypeTI 1:1:ROIMid 0.020860 0.073685 152.000000 0.283 0.777487
## StimTypeTI 1:3:ROIMid 0.085673 0.073685 152.000000 1.163 0.246776
## StimTypeTI 1:1:ROIPost -0.026089 0.073685 152.000000 -0.354 0.723779
## StimTypeTI 1:3:ROIPost 0.037827 0.073685 152.000000 0.513 0.608446
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) StTTI1:1 StTTI1:3 ROIMid ROIPst STTI1:1:ROIM STTI1:3:ROIM
## StmTypTI1:1 -0.520
## StmTypTI1:3 -0.520 0.500
## ROIMid -0.520 0.500 0.500
## ROIPost -0.520 0.500 0.500 0.500
## STTI1:1:ROIM 0.368 -0.707 -0.354 -0.707 -0.354
## STTI1:3:ROIM 0.368 -0.354 -0.707 -0.707 -0.354 0.500
## STTI1:1:ROIP 0.368 -0.707 -0.354 -0.354 -0.707 0.500 0.250
## STTI1:3:ROIP 0.368 -0.354 -0.707 -0.354 -0.707 0.250 0.500
## STTI1:1:ROIP
## StmTypTI1:1
## StmTypTI1:3
## ROIMid
## ROIPost
## STTI1:1:ROIM
## STTI1:3:ROIM
## STTI1:1:ROIP
## STTI1:3:ROIP 0.500

##
##
emmip(BA_seg, StimType~ROI)

```



```
estimated_means <- emmeans(BA_seg, ~ StimType*ROI)

# Plot
p.BA_seg_enc.L <- ggplot(Seg_Hip_data_enc.L, aes(x = ROI, y = BOLD, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.8, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold", hjust = 1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Ant","Mid", "Post"),
    labels=c("Ant","Mid", "Post")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size=14),
```

```

legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("Left Hippocampus") + scale_fill_manual(
ggsave(p.BA_seg_enc.L,file="Figure_3C.pdf", width = 10, height = 9, dpi = 300, units = "cm", device = "p
p.BA_seg_enc.L

```

